

Question:

What is the de Broglie wavelength of an electron with a kineticenergy of 1eV?

Answer Choices:

- A. 2.48×10^{-8} cm
- B. 5.92×10^7 cm/sec
- C. 8.66×10^{-6} cm
- D. 3.14×10^{-5} cm
- E. 9.81×10^{-4} cm
- F. 5.27×10^{-9} m
- G. 1.23×10^{-7} cm
- H. 4.14×10^{-10} m
- I. 6.625×10^{-27} erg sec
- J. 0.512×10^6 eV

Reasoning + Answer:

<think>

- Kinetic energy of electron , $KE = 1\text{ eV} = 1.6 \times 10^{-19}\text{ J}$.
- De Broglie wavelength , $\lambda = h / p$, where h is Planck's constant ($6.626 \times 10^{-34}\text{ Js}$) and p is momentum .
- Momentum , $p = \sqrt{2 m KE}$, where m is mass of electron ($9.11 \times 10^{-31}\text{ kg}$).
- Substituting , $\lambda = h / \sqrt{2 m KE}$.
- $\lambda = (6.626 \times 10^{-34}) / \sqrt{2 \times 9.11 \times 10^{-31} \times 1.6 \times 10^{-19}} = 2.48 \times 10^{-8}\text{ m} = 2.48 \times 10^{-8}\text{ cm}$.

</think>

<answer> $2.48 \times 10^{-8}\text{ cm}$ </answer>